

Madhavik Dogra

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Backend & Systems Engineer

EXPERIENCE

Gnorium

Founder & Software Development Engineer

Apr 2025 – Present

- Solo-architected and deployed a collaborative knowledge platform end-to-end – **RESTful APIs in Swift/Vapor**, **Neo4j** and **PostgreSQL** data layer, and **WebAssembly** frontend, all running on **AWS EC2/S3**.
- Designed a **4-stage data processing pipeline** (Ingest → Enrich → Define → Verify) with schema validation at each stage using **Pydantic**, ensuring data integrity across the full ingestion lifecycle.
- Chose **Neo4j** for graph-heavy relationship queries, **PostgreSQL** for transactional data, and **Swift over Python** for lower memory footprint and runtime costs; set up **zero-downtime blue-green deployments via GitHub Actions CI/CD**.
- Open-sourced **11 modular Swift packages** – **web-security** (auth/crypto), **diff-engine** (version control), **web-apis** (routing), **admin-core** (admin panel) – designed as composable libraries with clean dependency boundaries. Built the frontend in **Embedded Swift + WebAssembly**.

Nayan – traffic intelligence platform deployed across 12+ countries (TieCon 50 winner)

Software Engineer

Feb 2025 – Apr 2025

- Built **real-time data pipelines** ingesting live video streams from city-wide CCTV and dashcam networks; designed the backend to process, flag, and route anomalous plate detections to law enforcement APIs.
- Architected the **detect → track → classify** processing pipeline – coordinating multiple ML model stages, handling frame-level state, and streaming structured results to downstream services.
- Shipped Nayan's production **ANPR (Automatic Number Plate Recognition) service** – integrated detection and OCR stages into a single backend that serves **real-time surveillance streams across 12+ countries**.

TECHNICAL PROJECTS

Jobtain – Autonomous Job Application Agent | *Python, SQLAlchemy, Playwright, LangGraph* |

- Built an end-to-end agentic pipeline that takes a queue of job URLs, detects the ATS platform (**Workday**, **Greenhouse**, **Lever** via URL patterns + DOM fingerprinting), and autonomously fills and submits applications.
- Designed a **SQLAlchemy ORM layer** (SQLite) storing full candidate profiles – work history, education, projects, certifications – with a **precedence-based form-filling strategy**: DB fields → custom answers → LLM inference → human-in-the-loop escalation.

RAG Search Engine | *FastAPI, FAISS, SQLite, Gemini API* |

- Built a **retrieval API** over research papers – ingests documents into a **FAISS vector index**, serves queries via a search endpoint, and runs a validation layer that scores each response for factual relevance before returning it.

EDUCATION

Bennett University

B.Tech, Computer Science & Engineering (AI Specialization)

2024

TECHNICAL SKILLS

Languages: Python, Swift, C/C++, SQL

Backend & APIs: Swift/Vapor, FastAPI, RESTful API Design, SSE Streaming, Pydantic, SQLAlchemy ORM

Databases: PostgreSQL, Neo4j (Graph), SQLite, FAISS (Vector Search)

Infrastructure: AWS (EC2/S3), Docker, GitHub Actions CI/CD, Blue-Green Deployments, Playwright

CERTIFICATIONS

HarvardX: CS50x (Introduction to Computer Science), CS50AI

NVIDIA DLI: Accelerated Computing with CUDA Python, Building RAG Agents